KAVACH — On-device scam shield for Indian phone users

Arm Bharat AI-SoC Challenge 2026–27 · Problem Statement 3 (Cybersecurity & Digital Safety) · Phase 1 technical report

November 2026 · snapshot of 12 November 2026

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> **How to read this report.** It describes **one current build** (12 Nov 2026). Every number has a row in `docs/EVIDENCE_LEDGER.md` with its source file, sample size and status (reproduced on 12 Nov, measured once on the phone, observed, reported from an earlier build, or planned). Results from earlier builds that shaped the design are kept in §7 and marked as history.

1. Summary

KAVACH is **one Android app on one phone**. When a call comes from a number that is not in the contacts and the user puts it on speaker, KAVACH listens, recognises Hindi, Hinglish and English on the phone, and warns **during** the call. It joins what it hears with the SMS, links, app installs, screen sharing, payment-app use and network activity on the same phone, because Indian scams move across channels, and a Llama 3.2 1B model on the Arm CPU explains each warning in the caller's language. **All inference is on the phone**: the main app does not hold Android's `INTERNET` permission. The only things that can leave the phone are optional and user-enabled (an SMS to a family member, a Bluetooth link to a second KAVACH phone, and the separate Network Watch app's DNS forwarding; §6).

Current shipped configuration

Part	What the submitted build uses	Fallback / optional
Call audio	one phone; the call on speaker; Android's accessibility capture rule (§3.5)	second-phone Bluetooth mode; OEM system mode (untested on a retail phone)
Speech	Silero VAD → **AI4Bharat IndicConformer** (Hindi/Hinglish) and **NVIDIA Parakeet 110M** (English), int8 CTC, sentence by sentence; **Auto** language per sentence	small streaming Zipformer models if the big files are absent
Detector	cue lexicon (16 categories) + hashed classifier + kill chain, ≈ 4 ms per sentence	—
Explanation	Llama 3.2 1B + KAVACH LoRA, ExecuTorch 8da4w `.pte` (820 MB), Arm CPU	playbook text when Llama is missing, slow or invalid
Shield	notifications, Link Check, App Inspector, Screen-share Guard, call awareness	—
Network Watch	per-app traffic counters in the main app	separate DNS-watch / Wi-Fi-check app

Headline results (shipped detector, no Llama; re-run on 12 Nov 2026). Counts, with Wilson 95 % intervals where the sample is small:

Evaluation set	Scam calls reaching STOP	STOP before the money/OTP request	Genuine calls STOPped	Genuine calls at CAUTION
Synthetic test, phrasings never trained on (511 / 489)	**477/511 (93.4 %)**	420/511 (82.2 %)	**0/489** (CI 0–0.8 %)	29/489 (5.9 %)
Same, caller heard through the speaker (502 / 498)	375/502 (74.7 %)	304/502 (60.6 %)	**0/498**	26/498 (5.2 %)
Hand-written calls (16 / 14)	15/16	10/16	**0/14** (CI 0–21.5 %)	3/14
Locked held-out calls (10 / 10), never used for tuning	8/10 (CI 49–94 %)	5/10	**0/10** (CI 0–28 %)	4/10
Hindi, telephone band, IndicConformer (50 / 50, TTS voice)	29/50 (58 %)	22/50	**0/50**	3/50
Real cellular calls, one phone (2 scam / 1 genuine)	**2/2**	—	**0/1**	0/1

On the team's phone (Snapdragon 7s Gen 2): detector 4.07 ms per sentence; both speech models together ≈ 0.2 × real time; one minute of Auto listening drew 385 mA for the whole phone (screen on), which would be ≈ 7.6 % of the battery per hour if it held for an hour (a sustained run is planned, §7.8); Llama decodes 11.5–11.9 tokens/s.

The honest reading. No genuine call reached STOP in any set, but the samples of genuine calls written by people other than the generator are small (24 calls), so this does not establish a false-STOP rate below 2 %. The detector still misses 7–25 % of scam calls depending on the set, warns before the money step in 50–82 % of them, and raises CAUTION on 5–40 % of genuine calls. A new cross-channel ablation (§7.10) shows that

fusion adds STOPs at the risky action (an APK install or remote access during a call), but not earlier warnings, and that it can blame a genuine call for an unrelated scam SMS. Llama can only escalate a warning, never clear one (§3.2). The 20-call human pilot (`demo/pilot20/`) is the next real measure.

2. Problem and scope

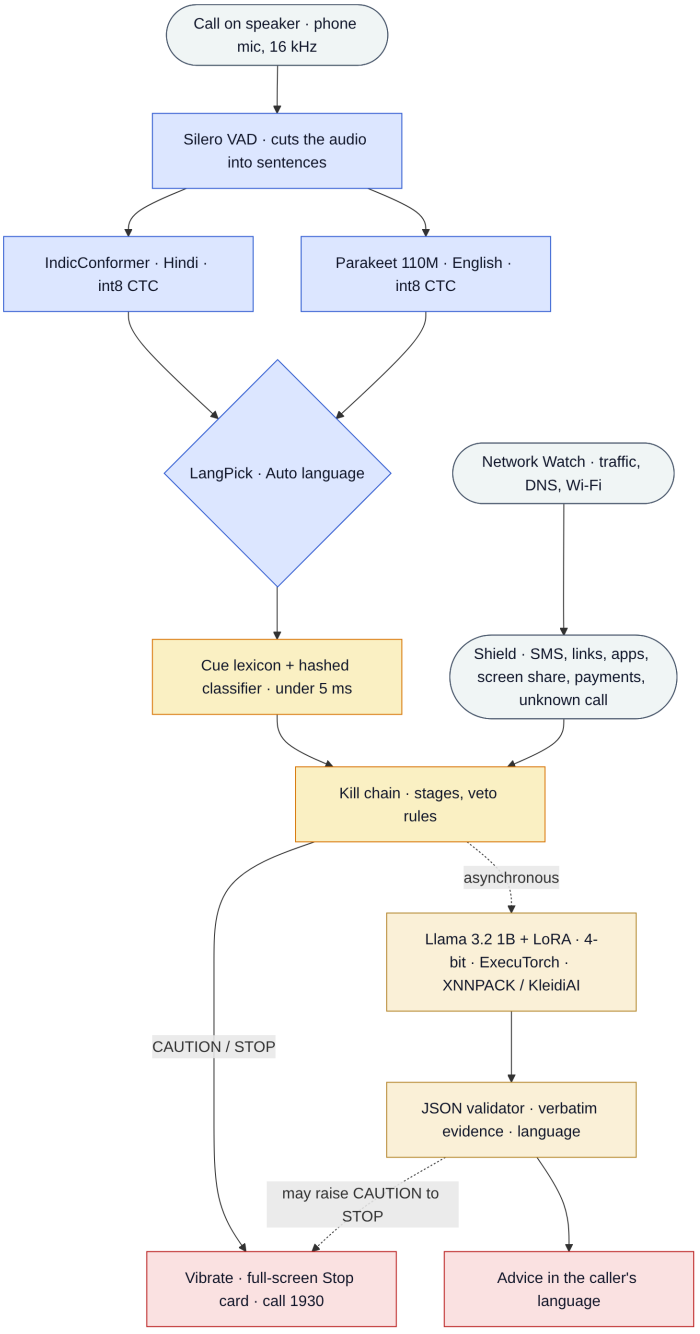
Indians lost about ₹22,845 crore to cyber fraud in 2024, about 206 % more than in 2023 (Ministry of Home Affairs reply in the Lok Sabha, July 2025). “Digital arrest” incidents on the National Cyber Crime Reporting Portal rose from 39,925 in 2022 to 1,23,672 in 2024, with reported losses of about ₹1,935 crore (Rajya Sabha reply, March 2025). A scam is a *sequence*: an authority claim, a threat or lure, isolation (“don’t disconnect, don’t tell your family”), an action (OTP, link, APK, screen share) and a payment. Single-channel filters see one fragment.

Problem Statement 3 asks for on-device AI that detects scam calls and messages, phishing links, malicious apps and network threats, explains them in plain language and recommends what to do, with a Llama-family model for the advisory part, run with ExecuTorch. KAVACH targets the mid-range Android phones most families own.

Since Android 10 ordinary apps cannot record a call. KAVACH listens through the documented accessibility capture exception while the call is on **speaker** (§3.5). The costs and what we do:

Cost	What KAVACH does	Section
The call must be on speaker	the opt-in KAVACH Phone dialer turns the speaker on for unknown numbers after a spoken notice; Shield’s call awareness still watches links, apps and payments if the user does not	3.4, 3.5
Android can silently stop the accessibility service	the app checks that the service is really running and that its recording is not silenced, and shows a Fix button	3.5, 7.5
Noisy rooms	larger speech models, a placement check, victim-side cues, and fusion with actions that need no audio	3.1, 7.3

3. Architecture



Current architecture: call listening (VAD → IndicConformer + Parakeet → LangPick), Shield and Network Watch feed one kill chain; Llama explains

3.1 Speech front-end (Arm CPU)

Stage	Implementation	Notes
Capture	`AudioRecord`, `VOICE_RECOGNITION` source, 16 kHz mono, 100 ms chunks	gain control and noise suppression off; audio held in memory only
VAD	Silero VAD (sherpa-onnx), 512-sample windows	cuts sentences: 0.5 s minimum silence, 12 s maximum
Hindi / Hinglish	AI4Bharat IndicConformer Hindi, 120 M parameters, int8 CTC (188.5 MiB)	Devanagari, including English loan words
English	NVIDIA Parakeet TDT-CTC 110M, int8 CTC (125.6 MiB)	punctuated English
Auto language	both models on every sentence; `LangPick` keeps the transcript whose words are more often real words of its language (wordfreq lists); borderline sentences keep the previous language; warnings follow the majority of the last 3 sentences	§7.4
Fallback	streaming Zipformer2-CTC Hindi (65.8 MiB) and Zipformer English (181 MiB)	only when the big files are missing
Placement check	SNR estimate from 20 ms frame powers over 5 s	“good / move closer / too noisy”
Own-voice filter	sentences heard while KAVACH speaks, and 1.2 s after, are ignored (and counted in the session log)	warnings are never spoken during a real call

3.2 Llama advisor (Llama 3.2 1B, ExecuTorch, Arm CPU)

Item	Choice
Model	Llama-3.2-1B-Instruct + KAVACH LoRA (r = 16, all projections), merged before export
Runtime	ExecuTorch `LlmModule` (Android AAR), XNNPACK delegate (KleidiAI micro-kernels are part of XNNPACK's Arm build; their use is not separately observed on the phone yet)
Quantisation	8da4w: 4-bit groupwise weights (g = 32), 8-bit dynamic activations, 4-bit embeddings; 820 MB
When it runs	only after the detector raises CAUTION or STOP, on its own thread; the rule-based warning is shown first
Latency	**warm prefix** (the constant system prompt pre-filled into the KV cache at start and re-armed after every alert); **early stop** when the JSON object closes; greedy decoding
Timeout	**35 s per generate call** (a re-prompt after invalid JSON gets its own). It covers waiting for a background warm-prefix prefill, the prompt prefill and the generation; not model loading. Set from the phone measurements: warm path ≈ 15 s, cold path of the fine-tuned file ≈ 26 s. It was 25 s in the build measured on 12 Nov; earlier versions of this report wrongly said 8 s. Self-test 60 s, benchmark 90 s
Contract	one JSON object: `scam, type, stage, confidence, evidence[≤3 verbatim quotes], say(≤35 words), action`
Validator	enums enforced, confidence clamped, quotes not found verbatim in the transcript dropped (confidence capped), unfinished last sentence cut, one re-prompt on invalid output; advice copied from the prompt's example or in the wrong language is replaced by playbook text and labelled
Authority	may raise CAUTION → STOP when its *self-reported* confidence ≥ 0.8 and stage ≥ isolation (this threshold is not calibrated); **never lowers or clears a warning**; invalid, timed-out or missing output leaves the rule-based warning unchanged

3.3 The always-on detector

1. **Cue lexicon** (`assets/cues.json`): 698 phrases in 16 categories (police and bank authority, identity misuse, threat, urgency, isolation, credential request, remote access, link, payment, reward, family emergency, safety advice, and three **victim-side** categories: reading out an OTP, saying they are paying, begging not to be arrested), each in English, romanised Hinglish and Devanagari, plus 208 mixed-script variants mined from what the Hindi model writes. Phrase rules allow gaps of 0–4 words; a request next to a negation ("never share your OTP") is safety advice.
2. **Hashed linear classifiers** (`cue_model`, `sms_model`, `url_model`): FNV-1a hashed words, word pairs and 3–5 letter fragments plus cue categories; ≈ 1 MB each; identical in Python and Kotlin; every score decomposes into per-word evidence.
3. **Kill chain** (`killchain.py` / `KillChain.kt`): evidence fades with a channel half-life (10 min calls, 30 min SMS and links, 24 h apps); the level is sticky within a session. STOP rules: isolation + authority; authority + identity + threat; two different action/payment requests; hook + action with no safety advice; victim complying with a scam pretext; suspicious call + risky action; unknown-number call + dangerous link /

APK / remote access / upload; two message channels together; total score. CAUTION: score, a payment app in an unknown call longer than 3 min, authority + hook. The rules use **time only**: they do not compare the organisation a caller claims with the sender of a message (§7.10).

4. **Guardian logic**: ticks after every sentence and at least every 3 s; the classifier adds CAUTION/STOP above 0.8 on two ticks in a row and can veto a weak rule-only STOP while it is confidently benign. Thresholds were tuned on the development sets only.

3.4 Shield (same app)

- **Message Shield** reads SMS / WhatsApp / Telegram *notifications* (no SMS permission) and runs the SMS classifier, the cues and Link Check on every URL. Notification text is evidence only: it is never given to Llama as an instruction.
- **Link Check**: an offline allow-list of genuine Indian domains (including `.bank.in`), look-alike rules (brand in an unofficial domain, typosquats, homoglyphs/punycode, shorteners, raw IPs, `@`, APK downloads, scam-prone TLDs, free site builders), then the URL model (threshold for ≤ 3 % false alarms on dev).
- **Call awareness**: knows that a call with an unknown number is on and for how long, without hearing it.
- **App Inspector v0** (preview): scores a new app's permission profile and sideloading. **Screen-share Guard v0** (preview): STOP when a payment app comes to the front while a remote-access app is *running*. An installed remote-access app alone is not a warning. A warning is advice; KAVACH never blocks a payment.
- **STOP card**: full screen, a plain reason, a one-tap call to 1930; an optional SMS to one family member.
- **KAVACH Phone** (opt-in: the user makes KAVACH the phone's call app). A complete phone app: Keypad (pressing Call with nothing typed fills in the last number), Recents from the phone's own call log, Contacts with search, and a call screen with Answer / Decline, mute, speaker, keypad tones, hold, a call timer, the caller's name, a SIM choice on dual-SIM phones, call waiting and screen-off at the ear. Incoming calls also raise a call notification with Answer / Decline and a full-screen intent, because some phones (Xiaomi's "pop-up windows" rule, Android 14's full-screen permission) block a background app from opening the call screen; the Setup tab checks both and opens the right setting. For an unknown number it says it is turning on the speaker, then does so; it shows the live KAVACH level and, on STOP, a one-tap End call. Telecom still rings and writes the call log. History: in the first version, calls rang but could not be answered when KAVACH was the phone app (no call notification, and the call screen was blocked). Status: verified on the team's phone on 12 Nov (Xiaomi, Android 15): answering from the call screen and from the notification with the screen locked, declining, Recents and Contacts all worked; hold and a dual-SIM choice were not separately tried.
- **Second-phone link** (optional): Bluetooth RFCOMM with Android pairing, authentication and encryption, paired devices only, an 8 KB message cap, and only forwarded event kinds accepted with clamped weights (hardened on 12 Nov).

3.5 Listening on the same phone

During a call Android gives the microphone to the call and silences other recorders; an enabled accessibility service is the documented exception. `CallListenAccessService` reads no screen content and performs no actions. With **Protect my calls** on, listening starts when an unknown-number call starts. During a real call the warning is not spoken (it would go into the call and block the listening): the phone vibrates and the STOP card appears.

On 12 Nov changing the default dialer made Android stop KAVACH and mark this service as crashed while its switch still showed "on"; every recording during a call was then silenced. The app now compares the enabled setting with the services Android actually runs and checks `AudioRecordingConfiguration.isClientSilenced()` during a call; both states appear on the Readiness screen with a Fix button. This fix is verified on the one phone; other phone makers are not tested. Play-store

distribution would need a declared accessibility use case or OEM system mode (`docs/threat_model.md`, `docs/oem_mode.md`).

3.6 Network Watch

- **App traffic watch** in the main app (no `INTERNET`): `NetworkStatsManager` per-app byte counters every 30 s; only suspect apps (sideloaded, new, remote-access, rated suspicious) are judged; `suspicious_upload` and `beaconing_app`. Detection times below come from simulated 30 s polling; the real counter granularity on the phone is not yet checked. A sideloaded backup or sync app is flagged by design and can be trusted with "I know this app".
- **Optional DNS-watch app** (`io.kavach.netwatch`, holds `INTERNET`): a local DNS-only `VpnService` learns which app looks up which name, forwards the look-up unchanged, and flags bank look-alikes and bursts of non-existent machine-generated names. It can miss look-ups made through Private DNS or an app's own DNS-over-HTTPS, and it attributes names to apps only as well as Android reports them.
- **Home Wi-Fi check**: connects to 10 commonly abused services on the local /24 of a network the user owns. An open port is reported as a risk, not as malware.
- **Flow model** (UNSW-NB15): evaluated, **raises no alerts** (§7.6).

4. Arm-specific engineering

Lever	Status	Evidence
ExecuTorch + XNNPACK `.pte`	fine-tuned 8da4w file runs on the phone; export verified token by token on a tiny model	`llama_pipeline_check.json`, phone screenshots
Quantisation	8da4w g32 + 4-bit embeddings (820 MB) vs Meta SpinQuant INT4 (1,083 MB): similar decode speed on the phone (11.5 vs 11.6–11.9 tok/s, single runs)	device JSON, self-test
KleidiAI on vs off	build script ready (`bench/build_llama_runner.sh`), **not run**	planned
Threads	speech RTF measured at 2 and 4 threads (Hindi 0.153 → 0.092); the energy cost of 4 threads is **not measured**	device JSON
big.LITTLE	the app logs the core each thread last ran on; no placement claim is made yet	session CSV
Battery and heat	one-minute listening sample only; 30-min soak script ready (`bench/thermal_soak.sh`)	planned
Fast path	0.44 ms per tick on a laptop JVM (host baseline, not a LITTLE-core number); 4.07 ms per sentence on the phone, all cores	`kotlin_engine_host.json`, device JSON

`qmode: int8` failed to export on the GQA projections in ExecuTorch 1.5.1 and `torchao:8da8w` cannot be combined with XNNPACK (both reproduced on the tiny model), so the shipped file is 4-bit.

5. Data

Need	Source	Use
Call transcripts	synthetic generator: 11 scam scripts, 9 genuine scenarios, 47 utterance pools (English, Hinglish, Devanagari), ASR-style corruption and timing; paraphrased from I4C, RBI and bank advisories	train 8,000 / dev 600 / test 1,000
Hard negatives	bank alerts that say “never share OTP”, delivery OTPs, relatives asking for money, people discussing scams, genuine police, office talk about payments and screen sharing	every split
Held-out phrasing	template-disjoint splits (pools partitioned by item)	dev for selection, test reported
Caller through speaker	`test_far`: 3× ASR errors on the caller’s lines	reported only
Independent checks	30 hand-written calls (other author style); 20 locked held-out calls written before tuning	reported only
ASR text	88 English and 410 Hindi training calls through Piper TTS → channel → recogniser	classifier training
SMS	UCI SMS Spam Collection (CC BY 4.0) + 4,300 synthetic Indian messages	`sms_model`
URLs	Phishing.Database (MIT), top-10k domains, genuine Indian domains, 8k generated look-alikes	`url_model`
Network	UNSW-NB15, NSL-KDD, 25 DGA families + popular domains, simulated per-app traces	Network Watch
Llama	4,000 / 400 instruction pairs from the phone’s own prompt code	LoRA
Human calls	20-call locked pilot (`demo/pilot20/`), consented role-play	planned 21 Nov

No real victim audio or personal data is used or committed.

6. Engineering quality, privacy and security

- **One source of truth**: lexicon, playbook, domain lists and weights live in `assets/`.
- **Parity**: `android/core` checks 336 golden outputs from the Python reference (normalisation, hashing, lexicon, features, all three models, URL and message verdicts, kill-chain steps, prompts, JSON validation, full replays, Network Watch, LangPick). Porting found two real bugs.
- **Tests**: 38 pytest tests (one found that `"scam": "maybe"` was read as *not a scam*).
- **Fail-safe**: if Llama is missing, slow or invalid, the rule path still warns with playbook text.
- **Design**: dark mode is true black with grey cards and a champagne-gold accent (light mode: ivory and cobalt); every text colour pair passes the 4.5 : 1 contrast check (`docs/design_tokens.md`); risk is always

written in words, never shown by colour alone.

- **What can leave the phone** (full table in `docs/EVIDENCE_LEDGER.md`): nothing from the main app's inference (no `INTERNET`); an optional SMS to one family member; an optional Bluetooth link to a paired KAVACH phone; the optional Network Watch app's DNS forwarding. Audio is never written to storage; the session log keeps times, levels, rules, sentence languages and Llama speed (no transcript; Llama's own advice is kept when it was replaced, for review).
- **Exported components** are protected by platform permissions (`BIND_INCALL_SERVICE`, `BIND_NOTIFICATION_LISTENER_SERVICE`, `BIND_ACCESSIBILITY_SERVICE`); the in-app link uses a signature-level permission.

7. Results

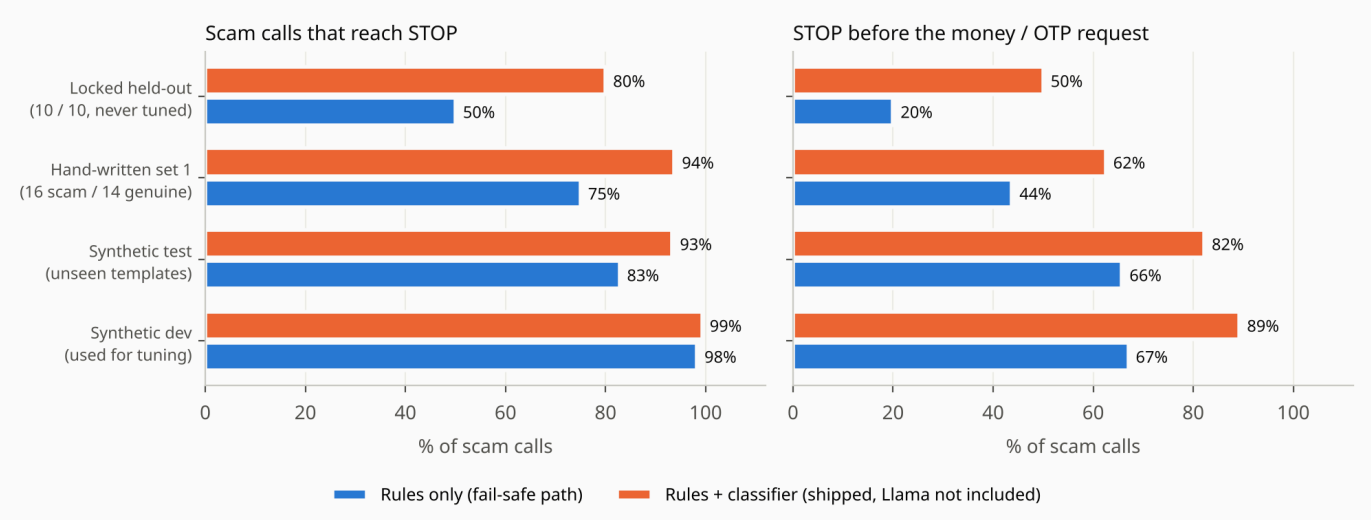
All host results below were re-run on 12 Nov 2026 with the shipped models unless marked otherwise.

7.1 Text classifiers (held-out test splits)

Model	Task	Test n	Precision	Recall	F1	ROC-AUC
`cue_model`	120-token call window → scam	9,010	0.946	0.883	**0.913**	0.962
`sms_model`	SMS / notification → scam	1,914	0.994	0.901	**0.945**	0.990
`url_model` (rules + model)	URL → phishing	10,196	0.987	0.752	0.853	0.945

Template-disjoint selection mattered: with a dev split that shared templates, the call model looked perfect on dev (F1 0.999) but scored 0.817 on test (history). **Link Check** flags 98.5 % of generated Indian look-alikes, 0 % of genuine Indian domains, 3.9 % of popular global domains and 69 % of generic global phishing domains.

7.2 Call-level detection (host, replayed in call time)



Call-level results

Set (scam / genuine)	Config	STOP	≥ CAUTION	STOP before money	Median time to STOP	False STOP	False CAUTION
Synthetic test (511 / 489)	rules only	423 (82.8 %)	472	336 (65.8 %)	17.3 s	0	0
	shipped	**477 (93.4 %)**	508 (99.4 %)	420 (82.2 %)	12.6 s	**0**	29 (5.9 %)
Caller through speaker (502 / 498)	rules only	312 (62.2 %)	360	218 (43.4 %)	18.9 s	0	0
	shipped	**375 (74.7 %)**	466 (92.8 %)	304 (60.6 %)	14.6 s	**0**	26 (5.2 %)
Hand-written (16 / 14)	rules only	12	14	7	24.7 s	2	3
	shipped	**15**	15	10	19.0 s	**0**	3
Locked held-out (10 / 10)	rules only	5	8	2	21.2 s	0	1
	shipped	**8**	8	5	21.3 s	**0**	4

The classifier veto removes both rule-only false STOPs on the hand-written set (people *discussing* a scam; an office call about screen sharing and payments). The 90 % “STOP before money” target is **not met** (50–82 %), and false CAUTIONs on the small human-written sets are high (3/14, 4/10). Earlier builds reported 90.6 % / 71.1 % / 70 % on the same sets; the gain comes from the Hindi lexicon and classifier changes of 11 Nov, which were made for Hindi phone failures, not tuned on these sets. The held-out set has been scored on several builds and is therefore no longer pristine.

7.3 Noisy rooms (host, earlier detector build)

Four scripted calls (2 scam, 2 genuine), Piper TTS, telephone channel, room reverb, fan/traffic hum, TV or babble at 30, 15, 8, 3 dB: 48 calls per configuration (`bench/e2e_noise.py`).

English recogniser + front-end	Scam STOP 30 / 15 / 8 / 3 dB	Genuine STOP	WER hum / TV / babble	Host RTF
20M (previous default)	3/6 · 3/6 · 0/6 · 0/6	0/24	0.53 / 0.80 / 0.74	0.05
20M + GTCRN speech denoiser	1/24 in total	0/24	0.48 / 0.97 / 1.13	+0.08
20M + scam-word hotwords	3/24 in total	0/24	0.60 / 0.74 / 0.75	
Zipformer en-2023-06-21	6/6 · 6/6 · 4/6 · 2/6	0/24	0.20 / 0.83 / 0.90	0.12

The larger model shipped (now the fallback behind Parakeet); the denoiser (invents speech from TV and babble) and hotwords (+1 call) were rejected. With background voices even the larger model transcribes the wrong speaker, which is why the app shows a placement check. One TTS realisation per condition: read these as a direction. Parakeet has not been run through this noise benchmark.

7.4 Hindi speech and Auto language

100 unseen synthetic Hindi calls (50 scam, 50 genuine), Piper TTS, decoded by the phone’s models:

Condition	Streaming Hindi model: red / yellow-or-red	IndicConformer: red / yellow-or-red	Genuine → red / yellow (IndicConformer)
near speaker, quiet room	44 % / 92 %	**66 % / 96 %**	0 % / 6 %
telephone band, SNR 10–30 dB	0 % / 46 %	**58 % / 86 %**	0 % / 6 %

The three genuine calls at yellow are a real traffic-police call and two real card-fraud checks. Lexicon and classifier fixes alone had moved telephone-band yellow-or-red from 12 % to 44 % with 0 % red; only the better recogniser made Hindi usable. All audio shares one TTS voice per language.

Auto language (held-out split, first two caller sentences of each call, `langpick_test.json`):

Channel	English chosen for English speech	Hindi chosen for Hindi speech
telephone band, SNR 10–30 dB	93.0 % (80/86)	91.0 % (91/100)
near speaker	87.2 % (75/86)	99.0 % (99/100)

Overall 92.7 % (345/372). How much a wrong choice changes the final warning is not yet measured.

7.5 On the phone

Phone: Xiaomi 2312DRA50I, Snapdragon 7s Gen 2 (SM7435: 4 × Cortex-A78 2.4 GHz + 4 × Cortex-A55 1.96 GHz), 7.3 GB RAM, **Android 15**, on battery.

Real cellular calls, one phone (12 Nov 2026, 03:15–03:53): 3/3 as expected (observed). A teammate called from a number not in the contacts; the call was answered in the phone's own dialer and put on speaker; KAVACH started by itself with the language on Auto.

Call	Result
Hindi digital-arrest scam (fake police, Aadhaar parcel, warrant, "tell no one", RBI safe account)	**STOP** during the call
English KYC scam (bank KYC, account blocked, SMS link, OTP)	**STOP** during the call; Auto chose English; type "Fake KYC / bank account block"
Genuine Hindi bank fraud check ("never share your OTP or PIN")	**no warning** for the whole call

Three calls by one caller are a demonstration, not a measurement; the 20-call pilot follows (§7.8).

Replayed test files (12 Nov, laptop speaker 10–30 cm): 6/6 as expected (observed). Hindi digital arrest red (twice); Hindi genuine bank call lowest hint only; English digital arrest red; English KYC red; English delivery and English genuine bank calls no warning. History: in an earlier build two English scams stopped at yellow because the *spoken* yellow warning muted the microphone for 19–33 s; yellow warnings are now silent.

Benchmark (measured once, 12 Nov 01:35, `bench/results/device/kavach_bench_20261112_013528.json`)

What	Result
Detector (rules + classifier + kill chain)	4.07 ms per sentence
Hindi IndicConformer (sentence)	RTF 0.153 at 2 threads, 0.092 at 4
English Parakeet 110M (sentence)	RTF 0.052 at 2 threads, 0.037 at 4
Small streaming fallback models	Hindi RTF 0.239, English 0.117 at 2 threads
Auto listening, **one minute** (VAD + both models, 2 threads each; screen on)	54 % of one core; **whole-phone** current 385 mA (≈ 7.6 % of 5,100 mAh per hour **if extrapolated**); 38.0 → 37.9 °C; thermal status 0; 235 MB
Llama 3.2 1B, **Meta SpinQuant INT4 file** (ExecuTorch, XNNPACK)	load 4.1 s, +1.1 GB; first token 12.9 s cold (768-token prompt) / 5.9 s warm prefix; complete answer 22.0 s cold / 15.1 s warm; prefill 56–60 tok/s; decode 11.6–11.9 tok/s; 858 mA while generating
Llama, **KAVACH LoRA 8da4w file** (self-test, observed)	verdict digital_arrest 0.91 with its own advice; first token 18.3 s cold; 11.5 tok/s

There is no idle baseline yet, so the listening current includes the screen and the rest of the phone.

7.6 Network Watch

Part	Test	Result
App traffic watch	**simulated** 2-hour traces, 200 per scenario, 30 s polling	spyware uploads, slow exfiltration, heartbeats: 100 % detected (uploads in 30–90 s, heartbeats in 10–35 min); chat, video, streaming, Play backup: 0 % false alarms; a sideloaded backup app is flagged by design
Domain model	6 DGA families never seen in training	cryptolocker 92 %, necurs 92 %; dictionary-word DGAs ≈ 0 % (blind spot); 0.36 % false alarms on 85k popular domains; 0 genuine Indian domains
Flow model, UNSW-NB15	82,332 test flows	P 0.939, R 0.885, F1 0.911
Flow model, NSL-KDD	KDDTest+ once	F1 0.854
Flow model on phone-like traffic	**replayed capture**	flags 60/60 normal HTTPS sessions (UNSW-NB15 has no normal TLS) → **raises no alerts**
Pcap replay demo	browsing + scan + DGA + look-alike + heartbeat + 2.4 MB upload	rules find all 5; none on browsing
Home Wi-Fi check	**emulator** network	254 addresses in 12 s; found the emulator's open ADB port

7.7 Llama 3.2 1B after LoRA fine-tuning (Colab, 12 Nov 2026)

LoRA r = 16 on all projections, 1 epoch over 4,000 pairs on a free T4 (4.1 h; training loss 0.111, dev loss 0.041), merged and exported with ExecuTorch 1.4. Scored with `models/llama/eval_llama.py` on the first 100 dev examples, **full-precision merged model** (`llama_lora_colab.json`):

Metric	Base Llama 3.2 1B	+ KAVACH LoRA
valid JSON verdict	71 %	**100 %**
scam / not scam correct	38 %	**89 %**
scam type correct	58 %	**76 %**
warning stage correct	71 %	**100 %**
quoted evidence really in the transcript	8 %	**96 %**
advice copied from the prompt's example	32 %	**0 %**
advice in the call's language	55 %	**100 %**

These examples come from the training generator and are not ASR text; the deployed 4-bit `.pte` has not been scored on an independent set (planned, §7.8). On the phone the 4-bit file wrote its own advice in English and Hindi (`docs/media/llama_phone.jpg`); its Hindi is understandable but not always polished, and one earlier Hindi run was replaced by playbook text (now logged).

7.8 Still to measure (scripts ready)

- **20-call human pilot** (`demo/pilot20/`, scoring `bench/pilot20_report.py`): frozen build, 10 scam / 10 genuine, 4 Hindi / 3 English / 3 Hinglish each, speaker-balanced, two independent annotators.
- Deployed 4-bit Llama on independent, ASR and prompt-injection prompts (`docs/OWNER_CHECKLIST.md`).
- 30-minute listening soak with an idle baseline; 2 vs 4 threads energy; KleidiAI on/off.
- Fusion with Llama escalation (configuration E of §7.10).
- Reliability: screen lock, backgrounding, missing/corrupt model, permission revoked, dialer change.
- KAVACH Phone: hold, call waiting and a dual-SIM call (basic answering is verified).

7.9 PS3 scope coverage

PS3 expectation	KAVACH component	Status
Scam / fraud call detection on device	call listening → detector → kill chain → Llama	built; synthetic, hand-written, held-out, TTS-audio sets; 3 real calls
Phishing SMS / messages	Message Shield	built, F1 0.945 (synthetic + UCI)
Malicious links	Link Check	built
Malicious / fake apps	App Inspector	v0 preview, not evaluated on malware data
Screen-sharing / remote-access fraud	Screen-share Guard + kill chain	v0 preview
Network threats	Network Watch	rules evaluated in simulation and replay; flow model research only
Cross-channel correlation	kill-chain hub	built; ablation in §7.10
On-device, privacy-preserving	no `INTERNET` in the main app; audio never stored	by construction; optional paths in §6
Explainable alerts in Indian languages	Hindi / English reasons, Llama explanation	observed on the phone; quality measured on dev data
Arm optimisation	ExecuTorch + XNNPACK, int8 ASR, 4-bit Llama	measured on one phone; ablations planned

7.10 Does cross-channel fusion help? (new, 12 Nov)

`bench/fusion_ablation.py` replays **identical** timestamped event traces under five configurations: **A** the call detector alone; **B** message / link / app detectors alone, with no knowledge of the call; **C** A and B combined independently (the higher level wins, no correlation); **B+** Shield with call awareness but no listening; **D** full fusion exactly as wired in the app (the listener gets the forwarded actions; Shield's kill chain gets every event, the call start/end and the listener's CAUTION/STOP; the user sees the higher of the two). **E** (D + Llama) is not run on the host. Calls come from the caller-through-speaker test set; SMS texts are 8 scam and 4 genuine team-written examples; 40 sessions per template; thresholds are the shipped ones (nothing tuned here); the plan was written in the script before the first run, and the dev split was used to debug the harness.

Scam template (40 each)	A STOP	C STOP	B+ STOP	**D STOP**	Warned before the risky step: C / D
scam call alone	37	37	0	**37**	31 / 31
+ scam SMS with a link mid-call	32	40	40	**40**	38 / 38
+ APK install	30	30	40	**40**	32 / 32
+ remote access, then a payment app	36	36	40	**40**	29 / 29
scam SMS 5 min before the call	30	37	33	**40**	37 / 37
first 40 % of the call, then an APK	7	7	40	**40**	26 / 26

Genuine template (40 each)	C STOP (tied to the call)	**D STOP (tied to the call)**	D ≥ CAUTION
genuine call alone	0 (0)	**0 (0)**	5
+ unrelated scam SMS mid-call	32 (0)	**40 (32)**	40
+ scam SMS 20 min after the call	32 (0)	**32 (3)**	40
+ payment app opened during the call	0 (0)	**4 (4)**	4
+ risky APK installed 2 days earlier	0 (0)	**0 (0)**	2
+ genuine bank SMS, duplicated and reordered	0 (0)	**0 (0)**	3
+ screen sharing for work, no payment	0 (0)	**40 (40)**	40

Reading. Fusion turns an ambiguous call into STOP **at the risky action** (APK install +10/40, remote access +4/40, a call heard only in part +33/40), but it gives **no earlier warning** than the independent combination in any template. Most of that gain comes from *knowing a call with an unknown number is on* (B+), not from what was heard; the speech adds +3/40 when a scam SMS arrived 5 min before the call. The costs are real: an unrelated scam SMS during a genuine call is blamed on the call (32/40), a genuine work screen-share during an unknown-number call is STOP (40/40) by design, and a payment app opened in a genuine call is STOP 4/40. The rules link events by time only; matching the organisation named on the call with the

message sender is the Phase 2 fix. These are synthetic traces replayed on the host; replay timing is not live Android latency.

8. Limitations and threat model

- **Speech is the dominant error source.** Human voices on speaker-phone calls are measured only by three real calls; all other audio uses one TTS voice per language. TV or other people talking near the phone is the hardest case.
- **Synthetic data**, written by the team; the hand-written and held-out sets reduce but do not remove author bias. Zero false STOPs on 24 human-written genuine calls does not establish a ≤ 2 % rate.
- **Before-payment warnings** (50–82 %) are below the 90 % target; false CAUTIONs on human-written genuine calls are frequent. Llama only escalates, so it cannot reduce false CAUTIONs.
- **Fusion is temporal only** (§7.10): it can blame a genuine call for an unrelated message.
- **Llama** is evaluated in full precision on generator data; the deployed 4-bit file is only observed. Its confidence is self-reported, not calibrated. Prompt-injection resistance rests on the contract (Llama cannot lower a warning; quotes must be verbatim), not on a measured attack success rate.
- **Adversaries** can avoid trigger words; the classifier, category combinations, Llama and actions that need no audio partly compensate.
- **Platform**: the call must be on speaker; one-phone listening relies on the accessibility capture rule and was tested on one Xiaomi phone; Play policy limits accessibility use (`docs/threat_model.md`).
- **Battery** is a one-minute sample; Network Watch numbers come from simulation, replay and an emulator.

9. Phase 2 plan

1. Human-recorded test set (100+ scam, 100+ genuine, consented), Hindi WER, recogniser adaptation.
2. Deployed-Llama evaluation (independent, ASR, injection prompts) and the Llama escalation effect (E).
3. Entity-aware fusion (organisation on the call vs message sender) to remove the §7.10 false blames.
4. Sustained energy (30–60 min with an idle baseline), 2 vs 4 threads, KleidiAI on/off, big vs LITTLE, a budget phone, and NPU offload of the speech encoders.
5. App Inspector on public malware data; flow model retrained on Android traffic; dictionary-DGA detection.
6. Consent-based pilot with 20–30 family volunteers; formative usability tests with 3–5 new users.

10. Reproducing everything

```
        pip install numpy scipy scikit-learn pytest           # core + tests
        ./data/public/download.sh                             # public datasets
        python data/synth/generate_calls.py && python data/synth/generate_sms.py && python
data/dev/build_handwritten.py
        python models/classifiers/train_text.py && python models/classifiers/train_url.py
        for s in dev test test_far handwritten heldout; do python bench/eval_calls.py --split $s -
-json bench/results/calls_$s.json; done
        python bench/fusion_ablation.py --split test_far --n 40                # §7.10
        python -m pytest -q tests && python tests/make_parity_vectors.py
        cd android && ./gradlew :core:test                     # Kotlin parity (336 checks)
        python models/netwatch/train_domain.py && python models/netwatch/train_ids.py
        python bench/netwatch_sim.py && python bench/netwatch_pcap_demo.py
        python bench/e2e_noise.py --asr zipformer-en-2023-06-21 --configs base hotwords
        scripts/build_apks.sh                                 # debug APKs (needs Android SDK)
        # Llama: notebooks/kavach_llama_lora_colab.ipynb on a Colab T4
```

Sources: Ministry of Home Affairs replies in Parliament (Lok Sabha, July 2025: cyber-fraud losses; Rajya Sabha, March 2025: digital-arrest incidents), as reported by Scroll and PTI/Inc42; RBI circular RBI/2025-26/28 (`.bank.in`); model and data licences in `docs/model_cards.md`, `docs/data_card.md` and `NOTICE.md`.